

PAPER

[View Article Online](#)
[View Journal](#) | [View Issue](#)Cite this: *J. Mater. Chem. A*, 2024, **12**, 32182Nanoconfinement-induced calcium ion redox charge storage of V_2CT_x MXene†Suman Yadav and Narendra Kurra *

Rechargeable calcium metal batteries are promising for sustainable stationary energy storage applications but are plagued by the irreversible nature of Ca metal anodes due to the formation of a non-ionically conductive solid electrolyte interphase. Alternatively, calcium-ion batteries require the development of charge storage hosts with the characteristics of reversibility and facile diffusivity of high charge density Ca-ions. We report on a foundational experimental study on vanadium carbide MXene (V_2CT_x) as a two-dimensional (2D) charge host for the reversible (de)intercalation of calcium-ions. Pseudocapacitance of V_2CT_x is induced by confining a room temperature ionic liquid (RTIL), 1-ethyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide (EMIM-TFSI), which helps in partial charge transfer across the electrode|electrolyte interface. Bode analysis reveals redox charge storage dynamics and corroborates well with a change in the oxidation state of vanadium upon insertion of calcium-ions, confirmed through *ex situ* X-ray photoelectron spectroscopy. V_2CT_x shows a reversible specific calcination capacity of 120 mA h g^{-1} at a current density of 50 mA g^{-1} . This work marks the primary experimental investigation on the feasibility of 2D MXene compositions for Ca-ion storage and provides plenty of opportunities for exploring nano-confined fluids for improved charge storage capacities in multivalent metal-ion electrolytes.

Received 22nd August 2024
Accepted 28th October 2024

DOI: 10.1039/d4ta05932h

rsc.li/materials-a

Introduction

The electrification of transportation and grid-scale sectors demands the development of sustainable energy storage solutions.¹ Commercialized Li-ion batteries (LIBs) show their dominance in the portable sector; however, lithium, due to its limited reserves may not be a viable option for large-scale energy storage applications.² Earth-abundant ionic charge carriers have turned out to be suitable alternatives for the development of low-cost energy storage devices in the post-LIB era.^{3–7} Notably, multivalent metal-ion charge carriers are able to transfer two (Zn^{2+} , Ca^{2+} , Mg^{2+}) or three (Al^{3+}) electrons per formula unit of charge storage hosts, possibly giving rise to higher capacities over monovalent metal-ions (Li^+ , Na^+ , K^+).⁸ Among multivalent metals, the promise of calcium is due to its high abundance (5th most abundant in the earth's crust) with a reduction potential (-2.87 V vs. standard hydrogen electrode, SHE) close to that of Li^+/Li (-3.04 V vs. SHE), not experiencing a large voltage penalty.⁹ Moreover, calcium-ions diffuse relatively faster across a charge storage host compared to other multivalent metal ions (Mg^{2+} , Zn^{2+} , and Al^{3+}) due to their low polarization strength.^{10,11} Therefore, significant efforts are being

put forth for the development of charge storage hosts for reversible (de)intercalation of calcium-ions.^{12,13}

Calcium metal itself is the best anode of choice based on its low cost and impressive theoretical volumetric capacity of $\sim 2073 \text{ mA h cm}^{-3}$ (gravimetric capacity of 1337 mA h g^{-1}).^{8,14} However, reversible plating and stripping of Ca is blocked by the formation of a nonporous solid electrolyte interphase (SEI) on Ca metal anodes.⁸ Specific electrolyte formulations were developed to realize reversible plating and stripping of Ca at room temperature.^{15,16} There are still debates on the dendrite formation and the safety risks of Ca metal batteries.¹⁷ In this scenario, many efforts have been focused on the development of robust charge storage hosts for Ca-ion batteries.¹⁸ Layered charge hosts such as graphite,^{19,20} TiS_2 ,²¹ V_2O_5 ,^{22–24} and other open channel materials such as Prussian blue analogues^{25–27} were studied for electrochemical storage of Ca-ions. However, (de)insertion of high charge density Ca-ions not only exhibits sluggish solid-state diffusivity but also leads to phase transformations with limited cycling stability.⁷ Large overpotentials may be required to drive the solid-state transportation of Ca-ions by overcoming the huge intercalation energy barriers.²⁸ Therefore, it is important to develop electrically conductive two-dimensional (2D) charge storage hosts with reversibility yet exhibiting redox storage characteristics against high charge density Ca-ions.

MXenes – two-dimensional (2D) transition metal carbides, nitrides, and carbonitrides – are a promising class of conductive

Department of Chemistry, Indian Institute of Technology Hyderabad, Kandi-502284, Sangareddy, Telangana, India. E-mail: narendra@chy.iith.ac.in

† Electronic supplementary information (ESI) available. See DOI: <https://doi.org/10.1039/d4ta05932h>

redox charge storage materials for multivalent metal-ion batteries.^{29,30} However, polar functional groups ($=O$, $-OH$, $-F$, and $-Cl$) introduced during the top-down wet chemical etching synthesis may exhibit strong binding with high charge density metal ions, leading to plausible irreversible parasitic reactions. Strategies such as pillaring and surface chemistry modifications of MXenes have been implemented to facilitate the reversible (de)intercalation of divalent metal ions such as Zn^{2+} and Mg^{2+} .^{31–33} Beyond $Ti_3C_2T_x$ MXene composition, V_2CT_x is yet another widely investigated electrode material for post LIBs.³⁴ V_2CT_x is a suitable material choice for energy storage since it is one of the lightest MXene compositions and has hydrophilic surface functional groups (T_x refers to functional groups such as $-F$, $-Cl$, $-OH$, and $=O$). Notably, V_2CT_x was also employed as a reversible charge storage host for aluminium batteries, and showed a specific capacity of 300 mA h g^{-1} at a discharge current density of 0.1 A g^{-1} .³⁵ To the best of our knowledge, there is no experimental study on employing V_2CT_x MXene or, in general, any MXene composition for the reversible electrochemical storage of Ca-ions despite promising theoretical predictions.³⁶ In general, solvated (or hydrated) metal ions form electrical double layers in the MXene interlayer spaces, giving rise to intercalation capacitance.³⁷ Here, we demonstrate a nanofluid confinement-induced pseudocapacitive charge storage of V_2CT_x MXene in a Ca-ion electrolyte. The kinetics of charge storage processes are analyzed by cyclic voltammetry, which corroborates well with the 3D Bode impedance analysis. *Ex situ* X-ray photoelectron spectroscopy (XPS) studies revealed the partial charge transfer between V_2CT_x MXene and Ca-ions, unlike the traditional formation of electrical double layers without charge transfer.

Results and discussion

Fig. 1a illustrates the synthesis of multi-layered V_2CT_x (ml- V_2CT_x) MXene by selective removal of the aluminium atomic

layers from a V_2AlC MAX precursor by a top-down wet chemical etching method (see the Experimental section). It was reported that concentrated HF etchant may introduce a high density of defects in V_2CT_x MXene when compared to a mixed acid etchant (HF/HCl) with a lower content of HF.³⁸ Moreover, the mixed acid method was initially employed for synthesizing $Ti_3C_2T_x$ MXene with minimal etching by-products (AlF_x , TiF_x).³⁹ The role of supporting acids such as HCl and H_2SO_4 was found to be the dissolution of etching by-products and their subsequent removal by washing *via* centrifugation. Similarly, in the synthesis of V_2CT_x MXene, it is presumed that etching by-products may be dissolved by an optimal choice of etching parameters of time and temperature.⁴⁰ Fig. 1b shows a typical secondary electron image of a dense layered V_2AlC MAX particle. After etching of interleaved 'Al' atomic layers, open and accordion-like morphology is clearly visualized for ml- V_2CT_x (Fig. 1c and S1, ESI†). The evolution of structural characteristics of ml- V_2CT_x was compared with respect to the parent V_2AlC MAX phase. The X-ray diffraction (XRD) pattern of the V_2AlC MAX phase shows a (002) peak at a 2θ value of 13.55° with a corresponding d -spacing of $6.53\text{ \AA}$ (Fig. 1d). The etching reaction cleaves the V-Al bonds, which results in unsaturated vanadium atomic sites in the lattice, causing the inevitable introduction of surface functional groups ($=O$, $-OH$, $-F$, and $-Cl$). Thus, the outcome of the etching reaction is the introduction of functional groups at the expense of removing the 'Al' atomic layer along with gas evolution, which leads to disruption of stacking order along the c -axis of V_2AlC MAX. These structural changes are evident from XRD patterns with a maximum intensity of the (002) peak relative to (013) for ml- V_2CT_x . Moreover, the downshift of the (002) peak for ml- V_2CT_x gives rise to an enlarged d -spacing of $5.5\text{ \AA}$ ($2\theta \sim 7.34^\circ$, corresponding to a d -spacing of $12.03\text{ \AA}$), which is in agreement with the literature reports.³⁸

To confirm the intrinsic quality of V_2CT_x MXene synthesized from the mixed acid etchant method, a bath-sonicated ml- V_2CT_x suspension was employed to record the ultraviolet (UV)-visible spectrum. As shown in Fig. 1e, the characteristic surface plasmon extinction peak at 296 nm corresponds to collective electron oscillations in the transverse direction and is in agreement with the reported literature.³⁸ Since delaminated V_2CT_x MXene is obtained by employing delaminating agents such as tetramethylammonium hydroxide (TMAOH) and tetrabutylammonium hydroxide (TBAOH), trapped large organic cations may be detrimental to the ease of movement of high-charge density cations in the interlayer spaces.³⁸ Thus, ml- V_2CT_x was employed as a charge storage host for studying the reversible electrochemical storage of Ca-ions. To facilitate the ease of transportation of Ca-ions through the interlayer spaces of ml- V_2CT_x MXene, the room temperature ionic liquid (RTIL) was pre-intercalated during the electrode fabrication process. 1-Ethyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide (EMIM-TFSI) was employed as a pre-intercalant due to its low vapor pressure and possibly act as a nanofluid for the ease of transportation of Ca-ions in the ml- V_2CT_x interlayer spaces.

The pre-intercalated ml- V_2CT_x MXene electrodes were fabricated by casting the active material consisting of ml- V_2CT_x , pre-

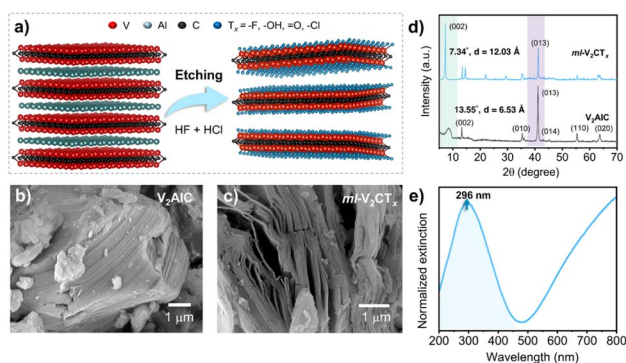


Fig. 1 Synthesis and characterization of ml- V_2CT_x MXene. (a) Schematic illustration of the synthesis of ml- V_2CT_x MXene from a V_2AlC MAX precursor *via* top-down wet chemical etching. Field emission scanning electron microscopy (FESEM) images of (b) V_2AlC MAX and (c) ml- V_2CT_x MXene. (d) XRD patterns of the V_2AlC MAX phase (grey) and ml- V_2CT_x MXene (blue). (e) UV-visible spectrum of ml- V_2CT_x MXene dispersed in water with a characteristic surface plasmon peak at 296 nm .

intercalated ionic liquid (IL), carbon black, and polyvinylidene fluoride (PVDF) binder onto a graphite foil current collector *via* a doctor blade coating method (see the Experimental section). The typical uniform thickness of the coated pre-intercalated ml- V_2CT_x MXene electrodes is $\sim 20\ \mu\text{m}$, as shown in Fig. 2a. The ionic liquid potentially occupies the sites of the micropores and surfaces through adsorption as well as intercalation into interlayer spaces. Fig. 2b shows the attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectra of the EMIM-TFSI ionic liquid (red) and the pre-intercalated ml- V_2CT_x MXene powder (vinous). The pure ionic liquid spectrum has a sharp absorption peak at $1048\ \text{cm}^{-1}$ (C-F stretching) in the anionic part (TFSI $^-$) and at $1574\ \text{cm}^{-1}$ (C=C stretching) in the cationic part (EMIM $^+$). The ionic liquid peaks become broad and are downshifted to $1042\ \text{cm}^{-1}$ (C-F stretching, TFSI anion) and $1566\ \text{cm}^{-1}$ (C=C stretching, EMIM cation) when intercalated into the ml- V_2CT_x MXene.⁴¹ Thermogravimetric analysis was performed to further characterize the thermal properties of pre-intercalated ml- V_2CT_x MXene (Fig. S2, ESI †). The bare ionic sizes of EMIM $^+$ and TFSI $^-$ are $7.74 \times 4.21\ \text{\AA}$ and $7.44 \times 4.41\ \text{\AA}$, respectively,⁴² lower than the *d*-spacing in ml- V_2CT_x MXene, possibly leading to insertion into interlayers (Fig. 2c). As there is a possibility of IL adsorption on the micropores and surface of the electrode besides the intercalation into interlayer spaces, minor contribution of adsorption processes may not be ruled out. The presence of EMIM-TFSI in the interlayer spaces of ml- V_2CT_x MXene was further confirmed by X-ray photoelectron spectroscopy (XPS), as shown in Fig. 2d. The survey scan of XPS confirms the presence of S2p and N1s peaks of EMIM-TFSI in the pre-intercalated V_2CT_x electrodes even after drying at $200\ ^\circ\text{C}$ under vacuum conditions for 12 h. The non-volatile nature of EMIM-TFSI helps it to remain in the liquid state confined into the interlayer space. Fig. 2e and f show the core level spectra of the S2p and N1s peaks at the binding energies of $168.2\ \text{eV}$ ($2p_{3/2}$) and $400.7\ \text{eV}$ (1s), respectively, which is in accordance with the

literature.⁴³ Moreover, XPS data of the free IL may be associated with uncertainties in the absolute binding energy values, and concrete analysis remains difficult in this situation.⁴³

Due to limitations associated with the reversibility of Ca metal,⁴⁴ the versatile electrochemical testing configuration was adapted to investigate the reversible electrochemical storage of Ca-ions across V_2CT_x MXene interlayer spaces. Over-capacitive activated carbon was employed as a counter electrode, with Ag wire as a quasi-reference electrode in a three-electrode configuration. The electrochemical behavior of the pre-intercalated ml- V_2CT_x MXene electrode was investigated in $0.8\ \text{M}$ calcium bis(trifluoromethanesulfonyl)imide ($\text{Ca}(\text{TFSI})_2$) electrolyte in a mixture of ethylene carbonate (EC), propylene carbonate (PC), and dimethyl carbonate (DMC) (3:3:4 vol%, respectively) solvents without any additive. The cyclic carbonates solvents (EC and PC) enhance the solubility of $\text{Ca}(\text{TFSI})_2$, whereas the linear carbonate solvent, *i.e.*, DMC, reduces the viscosity of the electrolyte as well as increasing the ionic conductivity.^{45,46} EC has film-forming properties at cathodic potentials and hence may promote desolvation at the electrode/electrolyte interface. Naked Ca-ion intercalation into V_2CT_x MXene could promote structural stability by suppressing the co-solvent intercalation.⁴⁷ Raman spectroscopy was employed to assess the coordination of the solvent molecules with the Ca-ion, as shown in Fig. S3, ESI † . The presence of a sharp peak at $780\ \text{cm}^{-1}$ (corresponding to free TFSI $^-$) in the Raman spectrum of $0.8\ \text{M}$ $\text{Ca}(\text{TFSI})_2/\text{EC}/\text{PC}/\text{DMC}$ electrolyte suggests the coordination of Ca-ions with the solvent molecules.⁴⁸

The stability of the Ag wire as a quasi-reference electrode was verified by taking the internal reference of the ferrocene redox system. Within the potential window of testing, the Ag wire reference electrode exhibited decent stability with a potential drift of $-2.44\ \text{mV}$ per hour (shown in Fig. S4, ESI †). Fig. 3a shows the cyclic voltammograms of the pristine (blue) and pre-intercalated (vinous) ml- V_2CT_x electrode at a scan rate of $0.3\ \text{mV s}^{-1}$. A pair of broad redox peaks are observed in the potential window of 0.0 to $0.5\ \text{V}$ (vs. Ag wire), which signifies the redox type of charge storage behavior of pre-intercalated ml- V_2CT_x electrodes in the Ca-ion electrolyte. To determine the role of the pre-intercalant, the electrochemical response of pristine ml- V_2CT_x (without pre-intercalation) was studied in the same Ca-ion electrolyte. Apparently, pristine ml- V_2CT_x showed rectangular-type CVs, which is an indication of the formation of electrical double layers (EDLs) across electrified interlayer spaces by the solvated Ca-ions. Such kind of intercalation capacitance is trivial in the case of MXene electrodes in an aprotic medium. Pre-intercalated ml- V_2CT_x MXene showed a typical capacity of $120\ \text{mA h g}^{-1}$ at a current density of $0.05\ \text{A g}^{-1}$, whereas the pristine ml- V_2CT_x electrode exhibited a specific capacity of $42\ \text{mA h g}^{-1}$ (Fig. 3b). Thus, the three-fold enhancement of charge storage capacity could be possibly attributed to the partial desolvation of Ca-ions by the confined EMIM-TFSI in the interlayer spaces of ml- V_2CT_x . By assuming V_2CO_2 as a composition, the theoretical calcination capacity is estimated to be $\sim 300\ \text{mA h g}^{-1}$. The deviation of experimentally realized capacity from the theoretical value could be due to inhomogeneous surface composition, which does not

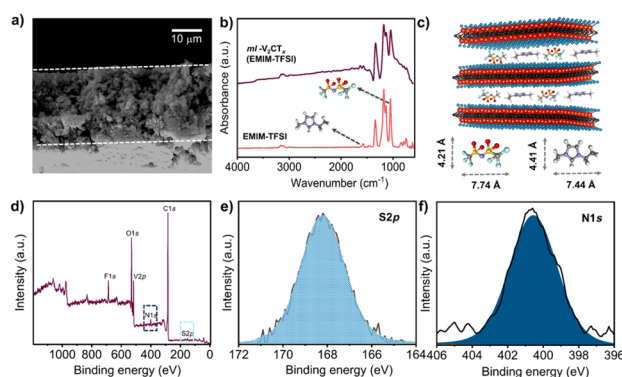


Fig. 2 V_2CT_x electrode fabrication and characterization. (a) Cross-sectional SEM image of the pre-intercalated ml- V_2CT_x MXene electrode coated over the graphite foil. (b) Attenuated total reflectance (ATR)-FTIR spectra of pure EMIM-TFSI ionic liquid (red) and pre-intercalated ml- V_2CT_x MXene (vinous). (c) Schematic representation of intercalated EMIM-TFSI in the interlayer of ml- V_2CT_x MXene. (d) XPS survey scan of the pre-intercalated ml- V_2CT_x electrode with characteristic core level spectra of (e) sulfur (S2p) and (f) nitrogen (N1s).

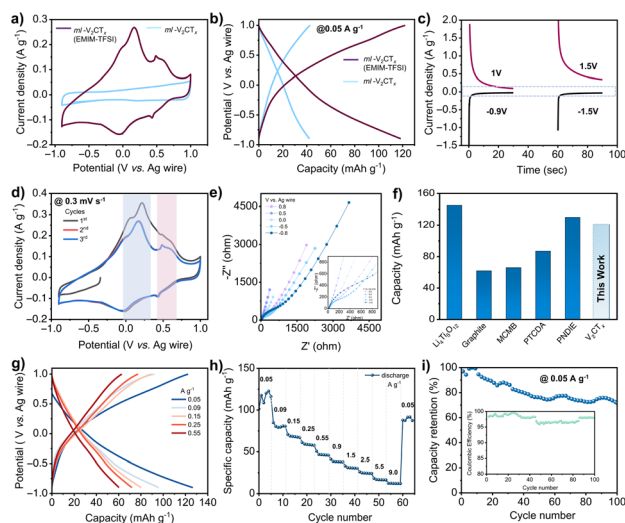


Fig. 3 Electrochemical behavior of pre-intercalated ml- V_2CT_x MXene in 0.8 M $Ca(TFSI)_2/EC/PC/DMC$ electrolyte. (a) Cyclic voltammograms of pristine (blue) and pre-intercalated (violet) ml- V_2CT_x MXene electrodes at a scan rate of 0.3 mV s^{-1} . (b) Galvanostatic charge-discharge (GCD) profiles at current densities of 0.05 A g^{-1} . (c) Chronoamperometry plot at different anodic and cathodic potentials. (d) Cyclic voltammograms of pre-intercalated ml- V_2CT_x MXene electrodes at a scan rate of 0.3 mV s^{-1} for 3 cycles after discharge at C/20 rate. (e) Electrochemical impedance spectra at different potentials (0.8 V, 0.5 V, 0.0 V, -0.5 V, and -0.8 V vs. Ag wire) in the 100 kHz to 10 mHz frequency regime with inset showing the high-frequency region. (f) Histograms showing the comparison of reversible calcination capacity of V_2CT_x MXene with previously reported electrode materials. (g) The charge-discharge profiles of pre-intercalated ml- V_2CT_x MXene at different current densities. (h) Rate performance of pre-intercalated ml- V_2CT_x MXene at different current densities. (i) Cycling performance of the pre-intercalated ml- V_2CT_x MXene electrode in non-aqueous 0.8 M $Ca(TFSI)_2$ electrolyte over 100 cycles at a current density of 0.05 A g^{-1} and inset showing the coulombic efficiency (%) with cycle number.

necessarily transfer two electrons per formula unit up on insertion of Ca-ions.

The electrochemical stable potential window of ml- V_2CT_x is found to be -0.9 V to 1.0 V (vs. Ag wire) in the $Ca(TFSI)_2/EC/PC/DMC$ electrolyte. The potential cut-offs of the electrolyte depend on the composition of the MXene electrode and the choice of solvent(s). For instance, $Ti_3C_2T_x$ MXene showed wider stable potential windows of 2.4 V (electrolyte: $LiTFSI-PC$)⁴⁹ and 2.7 V (electrolyte: EMIM-TFSI)⁴² in non-aqueous electrolytes in a versatile testing configuration. Chronoamperometry tests were employed to confirm the stable potential cut-offs of the V_2CT_x electrode in both cathodic and anodic potential regimes. A minor leakage current of $<0.1 \text{ A g}^{-1}$ was observed at -0.9 V (vs. Ag wire) and 1.0 V (vs. Ag wire), as shown in Fig. 3c. Leakage currents of above 0.1 A g^{-1} were observed by extending the potential limits to 1.5 and -1.5 V (vs. Ag wire), possibly due to irreversible side reactions. Moreover, a coulombic efficiency of 96% is achieved in the potential window of -0.9 to 1.0 V (vs. Ag wire). The pre-intercalated EMIM-TFSI ionic liquid in the interlayer spaces of V_2CT_x not only provides an easy movement for the Ca^{2+} ion to intercalate into the MXene galleries but also

increases the degree of confinement across the interlayer spaces. After the initial cycle of discharge at C/20 rate, the cyclic voltammogram (CV) test appears to start from an OCP of -0.4 V (vs. Ag wire) and drops to -0.9 V (vs. Ag wire) abruptly (shown in Fig. 3d). Subsequently, the second and third CV cycles reflect overlapping current profiles, as shown in Fig. 3d. Notably, redox peaks are observed for the pre-intercalated ml- V_2CT_x , unlike the case of pristine ml- V_2CT_x . A couple of anodic peaks at 0.15 V (vs. Ag wire) and 0.49 V (vs. Ag wire), with their corresponding cathodic peaks at -0.04 V (vs. Ag wire) and 0.44 V (vs. Ag wire) suggest specific potential dependent charge transfer interaction between Ca-ions and V_2CT_x MXene. However, pristine ml- V_2CT_x MXene showed (de)insertion of solvated Ca-ions in the entire potential window without charge transfer. Possibly, the confined EMIM-TFSI partially desolvates the intercalated Ca-ions and thus promotes charge transfer at specific potentials.

Electrochemical impedance spectroscopy (EIS) was employed to understand the kinetics of charge transfer between the pre-intercalated ml- V_2CT_x MXene and Ca-ion electrolyte at different applied dc potentials (-0.8, -0.5, 0.0, 0.5, and 0.8 V vs. Ag wire). The Nyquist plot provides information about the charge transfer kinetics, diffusion, and adsorption behavior of Ca-ions at the electrode|electrolyte interface. The impedance was recorded from high frequency (100 kHz) to low frequency (10 mHz), and distinct behavior was analyzed at different frequency regimes. Fig. 3e shows the Nyquist spectra at different applied dc potentials, i.e., 0.8 V, 0.5 V, 0.0 V, -0.5 V, and -0.8 V vs. Ag wire, where the intercept on the x-axis in the high-frequency region corresponds to equivalent series resistance (ESR) and is found to be 7Ω (inset, Fig. 3e). As the diameter of the semicircle in the high-frequency region changes with respect to applied dc potentials, charge transfer across the electrode|electrolyte interface is the contributing factor in this case.⁵⁰ Apparently, the diameter of the semicircle is lower at applied potentials of 0.0 V and 0.5 V (vs. Ag wire) compared to the rest of the applied dc potentials. This clearly indicates the facile charge transfer at specific potentials of 0.0 V and 0.5 V (vs. Ag wire), which is corroborated by the cyclic voltammograms, i.e., a pair of broad redox peaks are observed in the potential window of 0.0 V to 0.5 V (vs. Ag wire).

However, the Ca-ion storage capacity observed in pre-intercalated ml- V_2CT_x MXene is 2-fold higher than that of graphite¹⁹ ($62 \text{ mA h g}^{-1}@50 \text{ mA g}^{-1}$) and mesocarbon microbeads (MCMB)⁵¹ ($66 \text{ mA h g}^{-1}@2C$) and similar to that of lithium titanate oxide ($Li_4Ti_5O_{12}$)⁵² ($145 \text{ mA h g}^{-1}@35 \text{ mA g}^{-1}$) (Fig. 3f). Similarly, organic charge hosts such as 3,4,9,10-perylene-tetracarboxylic dianhydride (PTCDA),⁵³ polyimide poly[*N,N'*-(ethane-1,2-diyl)-1,4,5,8-naphthalenetetracarboxiimide] (PNDIE)⁵⁴ and poly(anthraquinonyl sulfide) (PAQS)@CNT⁵⁵ showed calcination capacities of $87 \text{ mA h g}^{-1}@20 \text{ mA g}^{-1}$, $130 \text{ mA h g}^{-1}@925 \text{ mA g}^{-1}$ and $116 \text{ mA h g}^{-1}@0.05 \text{ A g}^{-1}$, respectively, in saturated aqueous Ca-ion electrolytes. Fig. 3f plot is intended to compare the Ca-ion storage capacities despite differences in the charge storage mechanisms of carbon materials, such as graphite, MCMB, and lithium titanate oxide (LTO), and organic materials.

The charge–discharge profiles of pre-intercalated ml-V₂CT_x MXene at different current densities are shown in Fig. 3g (Fig. S5, ESI†). A typical sloping type profile with an initial discharge capacity of 120 mA h g^{−1} was observed at a current density of 0.05 A g^{−1}, which is 3 times higher than that of pristine ml-V₂CT_x MXene (Fig. 3b). The observed specific capacity of 120 mA h g^{−1} is of the active material consisting of ml-V₂CT_x MXene, pre-intercalated ionic liquid, carbon black, and PVDF binder. However, theoretical predictions estimated the Ca-ion storage capacity of 300 mA h g^{−1} for bare V₂C MXene.³⁶ But in reality, a reversible calcination capacity of 120 mA h g^{−1} is obtained for ml-V₂CT_x MXene. Such a discrepancy can be attributed to the inhomogeneous and random spatial distribution of functional groups on experimentally investigated V₂CT_x MXene. It is probable that the adsorption of Ca-ions on the V₂CT_x sheet may be influenced by the surface heterogeneity which further influences the insertion of desolvated ions into interlayer spaces. Nonetheless, enhancement in the calcination capacity is due to the confined EMIM-TFSI ionic liquid in the 2D layers of ml-V₂CT_x MXene that helps to partially desolvate and promote the interaction with the V₂C MXene lattice. The rate performance of the pre-intercalated ml-V₂CT_x MXene electrode at different current densities (from 0.05 A g^{−1} to 9 A g^{−1}) is shown in Fig. 3h, and the corresponding coulombic efficiencies are shown in Fig. S6a, ESI†. The electrodes showed a calcination capacity of 120 mA h g^{−1} at a current density of 0.05 A g^{−1} (at C/2.5), and one-fourth of the capacity was retained at a high current density of 0.15 A g^{−1} (50C), with a coulombic efficiency of 99%. After 100 charge–discharge cycles, 75% of capacity retention was observed at a current density of 0.05 A g^{−1} with coulombic efficiency of 99% (Fig. 3i). However, at high current density (9 A g^{−1}), the capacity retention is 54% after 1000 cycles with 100% coulombic efficiency (Fig. S6b, ESI†). A similar kind of capacity fading was observed for V₂CT_x MXene in Al-batteries.³⁵ The mechanism of capacity fading of MXene compositions in multivalent metal-ion electrolytes should be investigated in future studies.

The electrochemical kinetics of Ca²⁺ ions in the pre-intercalated ml-V₂CT_x MXene were investigated using scan rate-dependent voltammetry (Fig. 4). The CVs were recorded at different scan rates from 0.1 to 10 mV s^{−1} (Fig. S7, ESI†), and a reversible pair of anodic peaks with corresponding cathodic peaks suggests the redox-type charge storage of Ca-ions in V₂CT_x MXene (Fig. 4a). The CVs are presented specific capacitance (y-axis) to portray the prominence of broad redox peaks. Most importantly, specific capacity values are estimated and compared with the literature reports, as specific capacitance may not be appropriate in this context.²⁹ By considering the current (*i*)–scan rate (*ν*) power law relationship, the electrochemical kinetics of Ca-ions across V₂CT_x are analyzed based on eqn (1).

$$i = a\nu^b \quad (1)$$

where *a* and *b* are varying constants, and *i* is the current corresponding to scan rate (*ν*).

By taking logarithms on both sides, eqn (1) is changed to

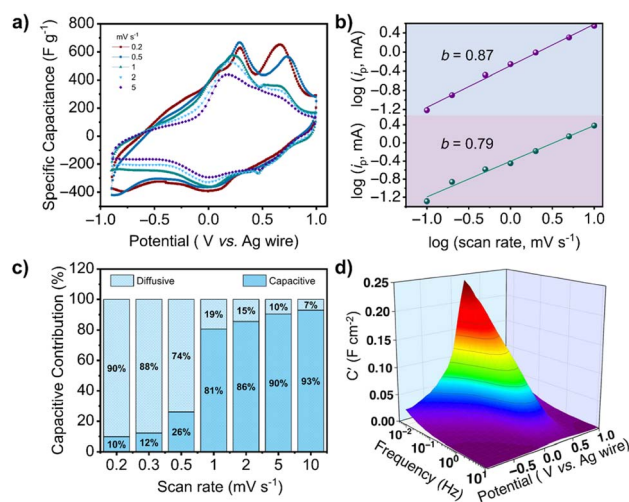


Fig. 4 Electrochemical kinetics of pre-intercalated ml-V₂CT_x MXene electrodes. (a) Cyclic voltammograms at various scan rates and (b) the corresponding log *i* versus log *ν* plot for *b* value analysis. (c) Histograms showing the capacitive and diffusive contributions at various scan rates. (d) 3D Bode plot of the real part of capacitance–frequency–potential of the pre-intercalated ml-V₂CT_x MXene electrode in 0.8 M Ca(TFSI)₂/EC/PC/DMC electrolyte.

$$\log i = \log a + b \log \nu \quad (2)$$

The slope of the log (anodic peak current) vs. log (scan rate) plot depicts the kinetics of ionic charge carriers across the electrode materials. When *b* = 1, the charge storage is dominated by a surface-controlled process, and a diffusion-controlled process is prevalent when *b* = 0.5. In this case, the *b* values are 0.9 and 0.8, corresponding to two anodic peaks at 0.15 V (vs. Ag wire) and 0.49 V (vs. Ag wire), respectively. The *b*-value of 0.8 signifies the hybrid type of charge storage kinetics with preference for surface-controlled processes over diffusion-limited ones (Fig. 4b).

The diffusive and capacitive contributions with respect to scan rates were calculated using *k*₁ & *k*₂ analysis by assuming total current is the sum of capacitive and diffusive currents

$$i(\text{total}) = i(\text{capacitive}) + i(\text{diffusive}) \quad (3)$$

$$i = k_1\nu + k_2\nu^{1/2} \quad (4)$$

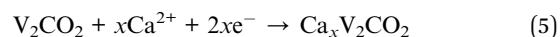
The total current response at a particular scan rate is a combination of both capacitive type and diffusive type (eqn (3)). The *k*₁*ν* and *k*₂*ν*^{1/2} in eqn (4) correspond to capacitive and diffusion contributions, respectively. At 0.2 mV s^{−1}, the capacitive contribution is 10% and gradually increases to 93% at 10 mV s^{−1} (Fig. 4c), signifying the dominant surface-controlled charge storage process at high scan rates. The cyclic voltammograms with the capacitive contribution at 1.0 and 5.0 mV s^{−1} scan rates are shown in Fig. S8, ESI†. To further understand the charge storage dynamics, V₂CT_x MXene galleries can be further characterized by constructing a 3D Bode plot of the real part of capacitance (*C'*) as a function of frequency (*f*) with a third independent variable, *i.e.*, potential (Fig. 4d). The real part of capacitance peaks at a potential of

0.2 V (vs. Ag wire) in the low-frequency region (1–0.01 Hz). At low frequencies, sufficient time (time (s) = 1/frequency) is available for the Ca-ions to intercalate into V_2CT_x interlayer spaces, possibly promoting the redox type of charge storage. The appearance of a waterfall kind of 3D Bode representation is a characteristic of pseudocapacitive type charge storage behavior. The Ca-ions at the rest of the potentials show a capacitive type of behavior, signifying the adsorption/intercalation of solvated Ca-ions across the interlayer spaces of V_2CT_x MXene.

To investigate the structural evolution on (de)insertion of Ca^{2+} ions across pre-intercalated ml- V_2CT_x MXene during charging/discharging cycles, *in situ* XRD was performed. Fig. 5a shows the corresponding charge/discharge profiles for the *in situ* XRD measurements, starting from OCP (potential V_1) to discharging to potential V_4 followed by charging to a potential of V_{12} . The successive second and third charge/discharge profiles are shown in Fig. S9, ESI†. After discharging and charging the pre-intercalated ml- V_2CT_x electrode, the (002) peak remained constant, *i.e.*, the *d*-spacing is 14.5 Å, as shown in Fig. 5b. Fig. 5c shows the corresponding heatmap of the (002) peak, showing no change in the *d*-spacing value during charging/discharging. Such kind of *in situ* XRD pattern reflects the feasibility of desolvated Ca-ion intercalation rather than co-solvent intercalation. In fact, it was shown that $Ti_3C_2T_x$ MXene in LiTFSI-PC electrolyte showed the same behavior.⁴⁹

Further, *ex situ* XPS was done to check the plausible interaction of Ca-ions with the V_2CT_x MXene functional groups. In the pristine pre-intercalated ml- V_2CT_x electrode, the V2p region was fitted into six components located at binding energy values of 515.7 eV ($V_{2p_{3/2}}$) and 522.7 eV ($V_{2p_{1/2}}$) for V^{3+} , 516.9 eV ($V_{2p_{3/2}}$) and 524.2 eV ($V_{2p_{1/2}}$) for V^{4+} , and 518.5 eV ($V_{2p_{3/2}}$) and 525.3 eV ($V_{2p_{1/2}}$) for V^{5+} , respectively (Fig. 5d). In a completely

discharged state (−0.9 V vs. Ag wire), the V^{4+} component reduces with the increase in the V^{3+} component, signifying the reduction of V^{4+} into V^{3+} upon insertion of Ca^{2+} ions, whereas there is no significant change in the V^{5+} component (Fig. 5e). Typically, V^{4+} to V^{3+} redox potentials are around 0.33 V (vs. SHE), and based on CVs shown in Fig. 4a, redox peaks were appeared within the potential window of 0.0 V to 0.5 V (vs. Ag wire). This clearly signifies the redox-active nature of the V^{4+} component, and the broadness is attributed to the inhomogeneous nature of surface states due to the random distribution of functional groups. Thus, reduction in the oxidation state of vanadium at the discharged state clearly reveals a redox type of charge storage in Ca-ion electrolyte. Despite the presence of Ca^{2+} in the charged electrode, the V^{4+} component increases at the expense of a decrease in the V^{3+} component, which suggests the extraction of Ca ions from the electrode during the charging process (Fig. 5f). The change of valence state of vanadium in V_2CT_x to accommodate calcium ions is given by the following general electrochemical reaction.



To further investigate the alteration in the structural morphology of the electrode material during electrochemical testing, *ex situ* SEM images were recorded (shown in Fig. S10, ESI†). Apparently, the morphology of the electrodes is maintained after cycling. Some of the fibers of the GF separator are noticed in the cycled electrodes (Fig. S10b, ESI†).

A few points are worth mentioning about this study. This is the first experimental investigation of reversible Ca-ion (de) intercalation into MXene materials. We have employed V_2CT_x as a model material, and it is well-known that MXene electrodes predominantly form electrical double layers in hydrated/solvated metal-ion electrolytes. Similarly, ml- V_2CT_x electrodes exhibit rectangular CVs due to the formation of EDLs by the solvated Ca-ions in the interlayer spaces, giving rise to limited charge storage capacities of 40 mA h g^{−1}. By introducing the ionic liquid as a pre-intercalant, the charge storage characteristics of ml- V_2CT_x changed to pseudocapacitive type with three-fold enhancement in the charge storage capacities. The pre-intercalated ml- V_2CT_x MXene has shown promising performance compared with the other reported electrode materials for Ca-ion storage (Fig. S11, Table S1, ESI†). *Ex situ* X-ray photoelectron spectroscopy analysis revealed that there could possibly be partial charge transfer between the desolvated Ca-ions and the MXene surface groups. Moreover, the screening effect provided by the ionic liquid helps in better diffusion of Ca-ions in the interlayer spaces in comparison with the pristine electrode. This study sheds light on how confinement-induced enhancement in the capacity and rate performance of MXene electrodes in multivalent metal-ions potentially advances the research on MXene materials in multivalent metal-ion-based energy storage devices.

Conclusions

We have demonstrated for the first time the use of vanadium carbide MXene (V_2CT_x) as a two-dimensional (2D) charge host for

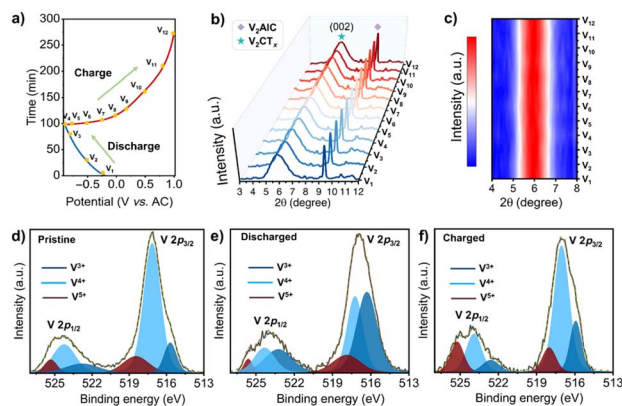


Fig. 5 (a) Galvanostatic charge–discharge (GCD) profiles of pre-intercalated ml- V_2CT_x MXene in 0.8 M $Ca(TFSI)_2$ during *in situ* X-ray diffraction (XRD) measurements. (b) *In situ* XRD patterns of pre-intercalated ml- V_2CT_x MXene at various charging/discharging potentials (discharging: −0.2 V, −0.5 V, −0.8 V, and −0.9 V (vs. AC), charging: −0.8 V, −0.5 V, −0.2 V, 0.0 V, 0.2 V, 0.5 V, 0.8 V, and 1.0 V (vs. AC)) in 0.8 M $Ca(TFSI)_2$ /EC/PC/DMC electrolyte. (c) Heatmap of the (002) peak, showing no change in the *d*-spacing value during charging/discharging. *Ex situ* high-resolution X-ray photoelectron spectra (XPS) of the V2p level in (d) the pristine state, (e) discharged state (−0.9 V vs. Ag wire), and (f) charged state (1.0 V vs. Ag wire).

reversible (de)intercalation of calcium ions. A nanoconfined room temperature ionic liquid (RTIL), 1-ethyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide, across V_2CT_x interlayer spaces enabled ease of transportation of high charge density Ca^{2+} ions. Interestingly, pseudocapacitive charge storage behavior was observed with a typical specific calcination capacity of 120 mA h g^{-1} at a current density of 0.05 A g^{-1} . *Ex situ* X-ray photoelectron spectroscopy confirmed the partial charge transfer between Ca-ions and the surface terminal groups of V_2CT_x . This study opens up new opportunities for exploring MXene compositions with plenty of scope for engineering interlayer spaces through nano-confined fluids for improved charge storage capacities and rate performance in multivalent metal-ion-based energy storage devices.

Experimental

Materials and methods

Synthesis of multi-layered V_2CT_x MXene. The multi-layered (ml) V_2CT_x MXene was synthesized from its V_2AlC MAX (particle size, 32 microns, Carbon Ukraine Ltd) precursor. The MAX precursor was etched *via* the mixed acid method, consists of 12 mL hydrofluoric acid (HF, 48 wt%, Acros Organics) and 8 mL hydrochloric acid (HCl, 12 M, 37 wt%, Fisher Chemical). While stirring at 400 rpm, 1 g of grey-colored V_2AlC MAX ceramic powder was gradually added to 20 mL of etchant solution. As the etching reaction is exothermic in nature, local heat production could result in the oxidation/dissolution of MAX rather than etching if not performed under ice bath conditions ($0\text{ }^\circ\text{C}$). Constant stirring of the above solution was continued until the formation of bubbles ceased. Afterward, the reaction mixture was shifted to an oil bath held at a temperature of $50\text{ }^\circ\text{C}$ while stirring at 400 rpm for 80 h. The etched acidic multi-layered V_2CT_x MXene was washed with deionized water several times *via* centrifugation at 3500 rpm for 5 minutes each time. Washing was continued until the pH reached ~ 6 . Typically, 1–1.5 L of deionized water is required for washing 1 g of multi-layered V_2CT_x MXene.

Fabrication of pre-intercalated V_2CT_x MXene electrode. The multi-layered V_2CT_x MXene sediment was dispersed in deionized water and filtered *via* vacuum-assisted filtration over a Celgard porous membrane (pore size, 60 nm). After draining the water present in the V_2CT_x MXene sediment, propylene carbonate (PC, Sigma Aldrich) solvent was run through the wet multi-layered V_2CT_x MXene. Thus, the trapped water in the V_2CT_x layers gets replaced by the PC solvent. This damp cake was soaked in 0.2 mL of neat 1-ethyl-3-methylimidazolium bis(trifluoromethyl sulfonyl)imide (EMIM-TFSI, Sigma Aldrich) ionic liquid (IL). The wet pre-intercalated multi-layered V_2CT_x cake was transferred to a desiccator by pouring an excess of 3–4 drops of neat EMIM-TFSI ionic liquid over it and kept under vacuum for 72 h at room temperature. The pre-intercalated multi-layer V_2CT_x MXene was vacuum-dried at $200\text{ }^\circ\text{C}$ overnight to eliminate the PC and trapped residual water. The electrode slurry was prepared by homogeneously mixing the pre-intercalated multi-layered V_2CT_x MXene (80 wt%), carbon black (Super P C645, 10 wt%, Timical), and polyvinylidene

fluoride (PVDF) binder (10 wt%, Sigma Aldrich) in a mortar and pestle. The mixture was ground until a smooth and thick slurry was obtained with the addition of a minimal amount of *N*-methyl-2-pyrrolidone (NMP). The slurry was coated over a graphitic foil current collector using a doctor blade coater, followed by vacuum drying at $70\text{ }^\circ\text{C}$ overnight, and punched into 3 mm diameter electrodes. Pre-intercalated electrodes consisting of four components – ml- V_2CT_x , pre-intercalated ionic liquid, conductive carbon black, and PVDF binder – had a typical areal mass loading of 1.4 mg cm^{-2} while the active material (including the weight of both ml- V_2CT_x and pre-intercalated ionic liquid) mass loading was 1.13 mg cm^{-2} . The typical electrode area of the pre-intercalated ml- V_2CT_x electrode was 0.07 cm^2 in the case of Swagelok test fixture and 0.78 cm^2 when using coin cell test fixture.

Preparation of activated carbon (AC) film. To prepare activated carbon (AC, YP50F grade) films, 95 wt% YP50F (Kuraray, Japan) was mixed and ground with 5 wt% polytetrafluoroethylene (60 wt% in water, Sigma Aldrich) in a mortar by adding a minimal amount of ethanol. The above slurry was rolled onto a film with a thickness of $\sim 100\text{ }\mu\text{m}$, followed by vacuum drying at $70\text{ }^\circ\text{C}$ for 12 h.

Material characterization. X-ray diffraction (XRD) patterns were recorded with a powder X-ray diffractometer (Malvern PANalytical, United Kingdom) with a $\text{Cu K}\alpha$ ($1.5406\text{ \AA}$) radiation source at a step size of 0.013° with 0.13 s dwelling time. The current of 40 mA and tension of 45 kV were maintained for recording the XRD pattern for all samples. The optical properties of the ml- V_2CT_x MXene dispersion were recorded using a JASCO model V-770 UV-Vis-NIR spectrophotometer. The morphology of the V_2AlC MAX precursor, ml- V_2CT_x MXene, and pre-intercalated ml- V_2CT_x was analyzed using scanning electron microscopy (SEM) (JEOL JIB4700F). The chemical states of elements in pre-intercalated ml- V_2CT_x MXene were identified using an X-ray photoelectron spectrometer (XPS) (AXIS SUPRA, Kratos Analytical, Germany) equipped with a monochromatic $\text{Al K}\alpha$ X-ray source ($h\nu = 1486.6\text{ eV}$). The data were analyzed with CasaXPS software, using linear background shapes. The interaction between the solvent and Ca ion was analyzed using Raman Spectroscopy. Raman spectra were recorded with a LabRAM HR Evolution Raman Spectrometer (Horiba Scientific, France) using a laser (532 nm , 10% laser power). The presence of EMIM-TFSI ionic liquid in the ml- V_2CT_x MXene was investigated by using Attenuated total reflectance Fourier Transform Infrared (ATR-FTIR) spectroscopy. The FTIR spectra were recorded using an ATR-FTIR (Alpha, Bruker) equipped with a diamond crystal, scanned from 500 to 4000 cm^{-1} at a resolution of 4 cm^{-1} . Thermogravimetric analysis was done by using TA Instruments thermal analyzer SDT Q600 (TA Instruments Pvt. Ltd, USA) from room temperature to $900\text{ }^\circ\text{C}$ at a heating rate of $10\text{ }^\circ\text{C min}^{-1}$ under an argon atmosphere with a 100 mL min^{-1} constant gas flow in an Al-type crucible.

Electrochemical measurements. The electrochemical measurements of V_2CT_x MXene as a working electrode were evaluated using an SP-150e electrochemical workstation (Bio-Logic, France). The electrochemical tests, such as cyclic voltammetry (CV), galvanostatic charge-discharge (GCD), and

electrochemical impedance spectroscopy (EIS), all were performed at 25 °C temperature. The electrochemical behavior of V_2CT_x MXene was studied *via* a three-electrode Swagelok cell assembly consisting of pre-intercalated ml- V_2CT_x MXene, activated carbon and Ag wire as the working, counter, and reference electrodes, respectively. The stability of the Ag wire as a quasi-reference electrode was verified by taking the internal reference of ferrocene. A 3-electrode cell consisting of activated carbon as the working and counter electrodes along with Ag wire as the reference electrode in the electrolyte mixture of 15 mM ferrocene + 0.8 M $Ca(TFSI)_2/EC/PC/DMC$ was employed for accessing the stability of Ag wire in the given electrolyte. Within the potential window of testing, the Ag wire reference electrode exhibited decent stability with a potential drift of -2.44 mV per hour.

The 3-electrode measurements were recorded in 0.8 M calcium bis(trifluoromethanesulphonyl)imide [$Ca(TFSI)_2$, TCI] electrolyte in ethylene carbonate (EC, Sigma Aldrich), propylene carbonate (PC, Sigma Aldrich), and dimethyl carbonate (DMC, Sigma Aldrich) solvents in a 3 : 3 : 4 vol% ratio, respectively. Glass microfiber (Whatman) membranes saturated with the electrolyte (roughly 0.1 mL) were employed as a separator between the working electrode and counter electrode. After cell assembly, it was kept under resting condition for 6 h in the OCP state. Initially, discharge was performed at C/20 rate from OCP (0.2 V *vs.* Ag wire) to -0.8 V *vs.* Ag wire. The cyclic voltammograms for pre-intercalated multi-layered V_2CT_x MXene were recorded at different scan rates (0.1–10 mV s $^{-1}$). Staircase potentiometric-electrochemical impedance spectroscopy (SPEIS) was used to investigate the charge transfer kinetics between the pre-intercalated ml- V_2CT_x MXene and Ca-ion electrolyte at different applied dc potentials (-0.8 , -0.5 , 0.0, 0.5, and 0.8 V *vs.* Ag wire). The potential at the working electrode was held for 10 minutes, and a small sinusoidal voltage amplitude of 10 mV was imposed on the pre-intercalated ml- V_2CT_x MXene electrode in the 100 kHz to 10 mHz frequency range.

Specific capacity was estimated by integrating current *versus* time plots.

$$\text{Specific capacity} = \frac{1}{m} \int idt \quad (6)$$

For estimating specific capacity, capacity was normalized with respect to the total weight of the electrode (including masses of ml- V_2CT_x , pre-intercalated ionic liquid, conductive carbon black and PVDF binder).

The real capacitance, C' , and imaginary capacitance, C'' , were calculated according to the following equations:

$$C' = -Z''/2\pi f|Z|^2 \quad (7)$$

$$C'' = Z'/2\pi f|Z|^2 \quad (8)$$

where $|Z|$ is the absolute value of the impedance (Ω), Z' and Z'' are the real and imaginary components of impedance, respectively, and f is the frequency (Hz).

The structural evolution during discharge/charge cycles was investigated using *in situ* XRD. The *in situ* XRD patterns were

recorded from 4° to 12° (2θ values) by using a powder X-ray diffractometer (Malvern PANalytical, United Kingdom) with a CuK_α (1.5406 Å) radiation source. For *in situ* measurements, a CR2032 coin cell having a 8 mm diameter hole in the positive terminal was assembled inside a glovebox ($O_2 < 0.1$ ppm and $H_2O < 0.1$ ppm). The coin cell consists of a pre-intercalated ml- V_2CT_x MXene film as a working electrode and an over-capacitive activated carbon as a counter electrode separated by glass microfiber (Whatman) membranes soaked in the Ca-ion electrolyte. A portable SP-150e electrochemical workstation (Bio-Logic, France) was used to apply the desired potentials on the working electrode during *in situ* XRD.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

SY acknowledges the scholarship support from PMRF, India. NK acknowledges support from the Indian Institute of Technology Hyderabad (Seed grant, SG-111), India and financial support from Science and Engineering Research Board, India (CRG/2021/001094).

Notes and references

- 1 D. Larcher and J. M. Tarascon, *Nat. Chem.*, 2015, **7**, 19–29.
- 2 G. Pistoia and B. Liaw, *Behaviour of Lithium-Ion Batteries in Electric Vehicles Battery Health, Performance, Safety, and Cost*, 1st edn, 2018.
- 3 J. W. Choi and D. Aurbach, *Nat. Rev. Mater.*, 2016, **1**, 16013.
- 4 M.-C. Lin, M. Gong, B. Lu, Y. Wu, D.-Y. Wang, M. Guan, M. Angell, C. Chen, J. Yang, B.-J. Hwang and H. Dai, *Nature*, 2015, **520**, 324–328.
- 5 A. Ponrouch, J. Bitenc, R. Dominko, N. Lindahl, P. Johansson and M. R. Palacin, *Energy Storage Mater.*, 2019, **20**, 253–262.
- 6 X. Zhang, H. Shi, L. Liu, C. Min, S. Liang, Z. Xu, Y. Xue, C. Hong and Z. Cai, *J. Colloid Interface Sci.*, 2022, **605**, 472–482.
- 7 P. Gao, H. Shi, T. Ma, S. Liang, Y. Xia, Z. Xu, S. Wang, C. Min and L. Liu, *ACS Appl. Mater. Interfaces*, 2021, **13**, 51028–51038.
- 8 Y. Liang, H. Dong, D. Aurbach and Y. Yao, *Nat. Energy*, 2020, **5**, 646–656.
- 9 A. Ponrouch, C. Frontera, F. Bardé and M. R. Palacin, *Nat. Mater.*, 2016, **15**, 169–172.
- 10 E. R. Nightingale Jr, *J. Phys. Chem.*, 1959, **63**, 1381–1387.
- 11 M. E. Arroyo-de Dompablo, A. Ponrouch, P. Johansson and M. R. Palacin, *Chem. Rev.*, 2020, **120**, 6331–6357.
- 12 B. Ji, H. He, W. Yao and Y. Tang, *Adv. Mater.*, 2021, **33**, 2005501.

- 13 L. Yan, W. Yang, H. Yu, L. Zhang and J. Shu, *Energy Storage Mater.*, 2023, **60**, 102822.
- 14 R. J. Gummow, G. Vamvounis, M. B. Kannan and Y. He, *Adv. Mater.*, 2018, **30**, 1801702.
- 15 D. Wang, X. Gao, Y. Chen, L. Jin, C. Kuss and P. G. Bruce, *Nat. Mater.*, 2018, **17**, 16–20.
- 16 Z. Li, O. Fuhr, M. Fichtner and Z. Zhao-Karger, *Energy Environ. Sci.*, 2019, **12**, 3496–3501.
- 17 S. D. Pu, C. Gong, X. Gao, Z. Ning, S. Yang, J.-J. Marie, B. Liu, R. A. House, G. O. Hartley, J. Luo, P. G. Bruce and A. W. Robertson, *ACS Energy Lett.*, 2020, **5**, 2283–2290.
- 18 X. Deng, L. Li, G. Zhang, X. Zhao, J. Hao, C. Han and B. Li, *Energy Storage Mater.*, 2022, **53**, 467–481.
- 19 S. J. Richard Prabakar, A. B. Ikhe, W. B. Park, K.-C. Chung, H. Park, K.-J. Kim, D. Ahn, J. S. Kwak, K.-S. Sohn and M. Pyo, *Adv. Sci.*, 2019, **6**, 1902129.
- 20 J. Park, Z.-L. Xu, G. Yoon, S. K. Park, J. Wang, H. Hyun, H. Park, J. Lim, Y.-J. Ko, Y. S. Yun and K. Kang, *Adv. Mater.*, 2020, **32**, 1904411.
- 21 S. Seong, H. Lee, S. Lee, P. M. Nogales, C. Lee, Y. Kim and S.-K. Jeong, *Batteries*, 2023, **9**, 500.
- 22 M. Hayashi, H. Arai, H. Ohtsuka and Y. Sakurai, *J. Power Sources*, 2003, **119–121**, 617–620.
- 23 X. Qin, X. Zhao, G. Zhang, Z. Wei, L. Li, X. Wang, C. Zhi, H. Li, C. Han and B. Li, *ACS Nano*, 2023, **17**, 12040–12051.
- 24 B. Jeon, H. H. Kwak and S.-T. Hong, *Chem. Mater.*, 2022, **34**, 1491–1498.
- 25 A. L. Lipson, B. Pan, S. H. Lapidus, C. Liao, J. T. Vaughey and B. J. Ingram, *Chem. Mater.*, 2015, **27**, 8442–8447.
- 26 N. Kuperman, P. Padigi, G. Goncher, D. Evans, J. Thiebies and R. Solanki, *J. Power Sources*, 2017, **342**, 414–418.
- 27 B. Wang, Y. Han, X. Wang, N. Bahlawane, H. Pan, M. Yan and Y. Jiang, *iScience*, 2018, **3**, 110–133.
- 28 A. Ponrouch, D. Tchitcheкова, C. Frontera, F. Bardé, M. E. A.-d. Dompablo and M. R. Palacín, *Electrochem. Commun.*, 2016, **66**, 75–78.
- 29 B. Anasori, M. R. Lukatskaya and Y. Gogotsi, *Nat. Rev. Mater.*, 2017, **2**, 16098.
- 30 F. Ming, H. Liang, G. Huang, Z. Bayhan and H. N. Alshareef, *Adv. Mater.*, 2021, **33**, 2004039.
- 31 M.-Q. Zhao, C. E. Ren, M. Alhabeb, B. Anasori, M. W. Barsoum and Y. Gogotsi, *ACS Appl. Energy Mater.*, 2019, **2**, 1572–1578.
- 32 M. Xu, S. Lei, J. Qi, Q. Dou, L. Liu, Y. Lu, Q. Huang, S. Shi and X. Yan, *ACS Nano*, 2018, **12**, 3733–3740.
- 33 M. S. Javed, A. Mateen, S. Ali, X. Zhang, I. Hussain, M. Imran, S. S. A. Shah and W. Han, *Small*, 2022, **18**, 2201989.
- 34 S. Behl, V. Lahariya, P. P. Pandey and R. Kumar, *J. Energy Storage*, 2024, **92**, 112176.
- 35 A. VahidMohammadi, A. Hadjikhani, S. Shahbazmohammadi and M. Beidaghi, *ACS Nano*, 2017, **11**, 11135–11144.
- 36 Y. Xie, Y. Dall'Agnese, M. Naguib, Y. Gogotsi, M. W. Barsoum, H. L. Zhuang and P. R. C. Kent, *ACS Nano*, 2014, **8**, 9606–9615.
- 37 S. Yadav and N. Kurra, *Energy Storage Mater.*, 2024, **65**, 103094.
- 38 K. Matthews, T. Zhang, C. E. Shuck, A. VahidMohammadi and Y. Gogotsi, *Chem. Mater.*, 2022, **34**, 499–509.
- 39 M. Anayee, N. Kurra, M. Alhabeb, M. Seredych, M. N. Hedhili, A.-H. Emwas, H. N. Alshareef, B. Anasori and Y. Gogotsi, *Chem. Commun.*, 2020, **56**, 6090–6093.
- 40 M. Seredych, C. E. Shuck, D. Pinto, M. Alhabeb, E. Precetti, G. Deysher, B. Anasori, N. Kurra and Y. Gogotsi, *Chem. Mater.*, 2019, **31**, 3324–3332.
- 41 F. W. Richey, B. Dyatkin, Y. Gogotsi and Y. A. Elabd, *J. Am. Chem. Soc.*, 2013, **135**, 12818–12826.
- 42 G. Valurouthu, R. Panigrahi, M. Saraf, C. E. Shuck, B. S. Mallik, N. Kurra and Y. Gogotsi, *Batteries Supercaps*, 2023, **6**, e202300009.
- 43 E. F. Smith, F. J. M. Rutten, I. J. Villar-Garcia, D. Briggs and P. Licence, *Langmuir*, 2006, **22**, 9386–9392.
- 44 R. Dugas, J. D. Forero-Saboya and A. Ponrouch, *Chem. Mater.*, 2019, **31**, 8613–8628.
- 45 T. Hou, K. D. Fong, J. Wang and K. A. Persson, *Chem. Sci.*, 2021, **12**, 14740–14751.
- 46 J. Li, C. Han, X. Ou and Y. Tang, *Angew. Chem., Int. Ed.*, 2022, **61**, e202116668.
- 47 H. Guo, M. Elmanzalawy, P. Sivakumar and S. Fleischmann, *Energy Environ. Sci.*, 2024, **17**, 2100–2116.
- 48 J. D. Forero-Saboya, E. Marchante, R. B. Araujo, D. Monti, P. Johansson and A. Ponrouch, *J. Phys. Chem. C*, 2019, **123**, 29524–29532.
- 49 X. Wang, T. S. Mathis, K. Li, Z. Lin, L. Vlcek, T. Torita, N. C. Osti, C. Hatter, P. Urbankowski, A. Sarycheva, M. Tyagi, E. Mamontov, P. Simon and Y. Gogotsi, *Nat. Energy*, 2019, **4**, 241–248.
- 50 T. S. Mathis, N. Kurra, X. Wang, D. Pinto, P. Simon and Y. Gogotsi, *Adv. Energy Mater.*, 2019, **9**, 1902007.
- 51 S. Wu, F. Zhang and Y. Tang, *Adv. Sci.*, 2018, **5**, 1701082.
- 52 C. H. Lee, C. S. Kim and S. K. Jeong, *Key Eng. Mater.*, 2017, **724**, 97–101.
- 53 I. A. Rodríguez-Pérez, Y. Yuan, C. Bommier, X. Wang, L. Ma, D. P. Leonard, M. M. Lerner, R. G. Carter, T. Wu, P. A. Greaney, J. Lu and X. Ji, *J. Am. Chem. Soc.*, 2017, **139**, 13031–13037.
- 54 S. Gheyhani, Y. Liang, F. Wu, Y. Jing, H. Dong, K. K. Rao, X. Chi, F. Fang and Y. Yao, *Adv. Sci.*, 2017, **4**, 1700465.
- 55 S. Zhang, Y. Zhu, D. Wang, C. Li, Y. Han, Z. Shi and S. Feng, *Adv. Sci.*, 2022, **9**, 2200397.